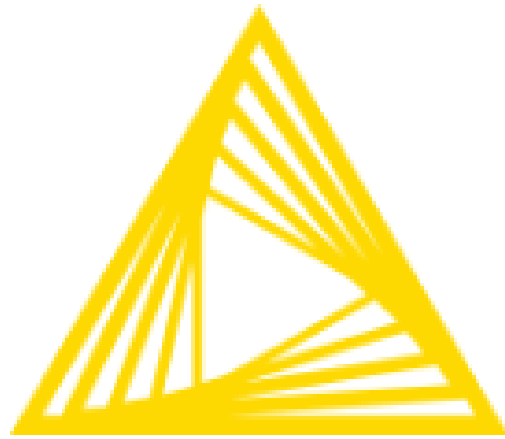


Data Preparation in action

KNIME workflows



Open for Innovation

KNIME

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• • • • • • • • • •

COS30045 – KNIME Analytics Platform

What is KNIME?

- Free, open-source data analytics program
- No-code drag and drop user interface

Why are we using KNIME?

- To help us process our data
 - Combine data files
 - Clean data, deal with missing values
 - Transform and summarise data
- **Not** going to use for modeling/machine learning

COS10022 Data Science Principles should already be familiar with KNIME



.
.

KNIME Workflow

A sequence of steps necessary to get to a specific result

Step 1

- Read Data

Step 2

- Clean Data

Step 3

- Filter Data

Step 4

- Train a model

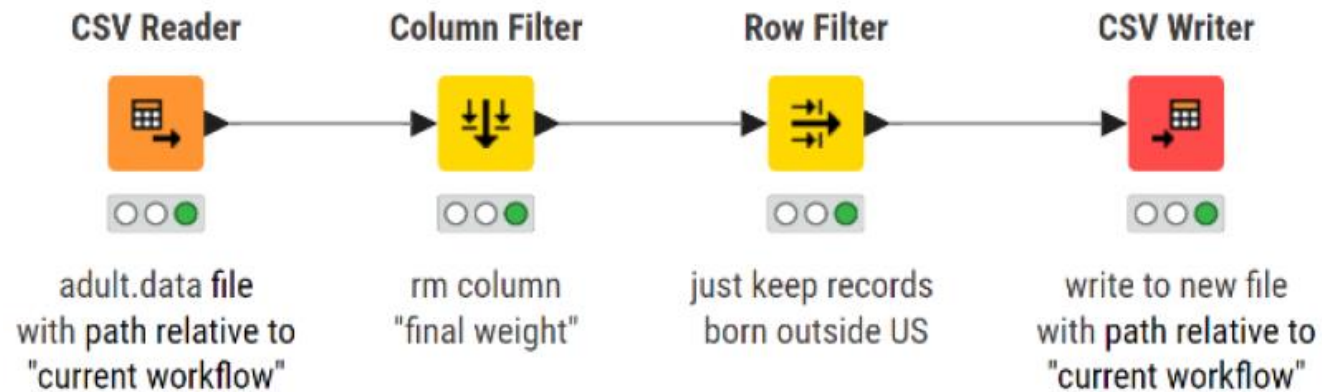


Figure 1.6. Example of a KNIME Workflow.

KNIME Workflow - Overview

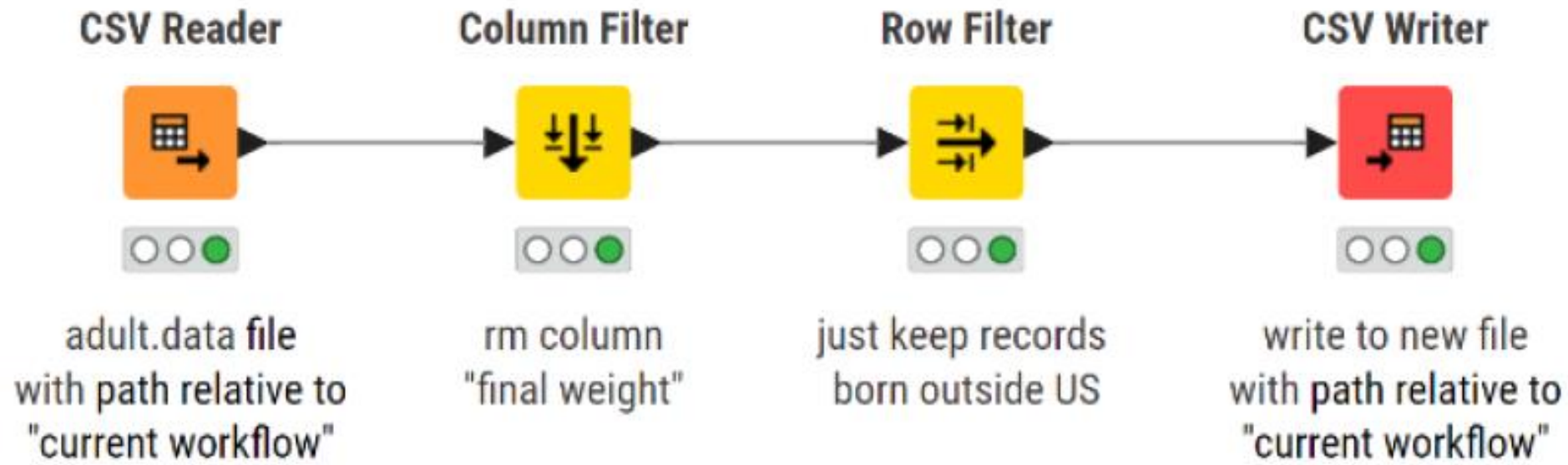


Figure 1.6. Example of a KNIME Workflow.

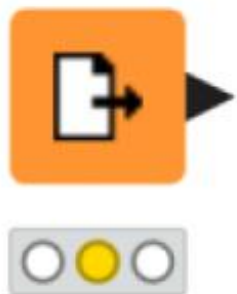
KNIME Workflow – Node status

File Reader



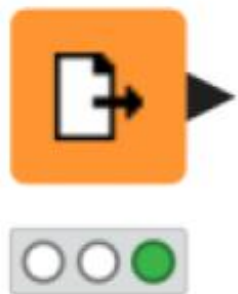
Inactive not yet configured

File Reader



Configured but not yet executed

File Reader



Executed successfully

File Reader



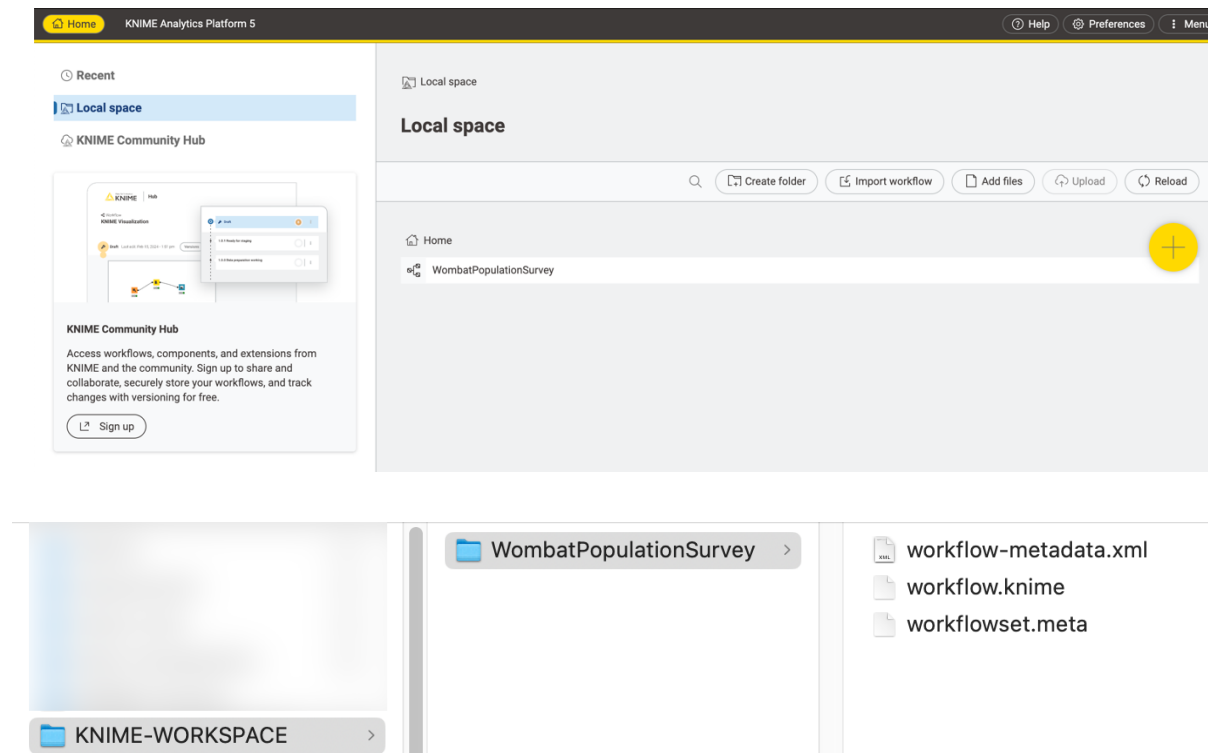
Executed with errors

.
.

Create KNIME local Workspace

Create KNIME workspace folder when you open it for first time

Items in KNIME workspace can be accessed in KNIME local space



<https://nre.tas.gov.au/wildlife-management/fauna-of-tasmania/mammals/possums-kangaroos-and-wombats/wombat>

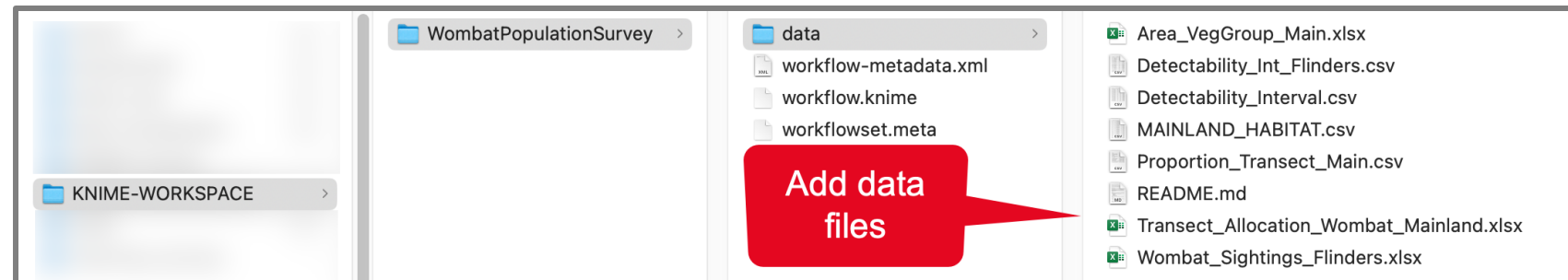
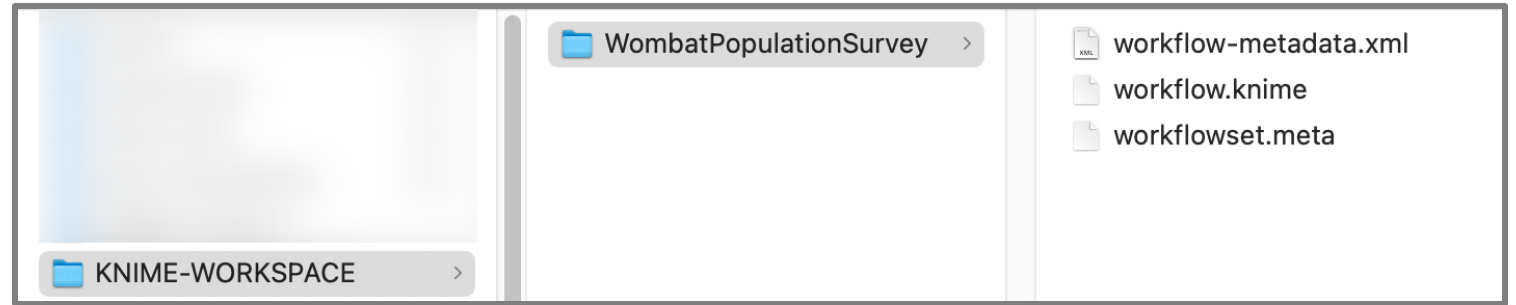
Read in data - data locations

Data files can be accessed in many ways...

Add data folder into your workflow folder:

- Gives you relative access to data
- Data is included in export files (eg .knwf file)

Use KNIME hub if swapping between computers



Understand data – Data dictionary

```
116 #####
117
118 DATA-SPECIFIC INFORMATION FOR: MAINLAND_HABITAT.csv
119
120 1. Short description: shows various data relating to Tasmanian mainland spotlight surveys
121
122 2. Number of variables: 11
123
124 3. Number of cases/rows: 19548
125
126 4. Variable list:
127
128 * Surveys.Survey_ID: individual ID assigned to each individual spotlight survey
129
130 * Dist_Min: minimum distance (in metres) of wombat sightings/counts according to previously set
131 distance intervals/bands
132
133 * Dist_Max: maximum distance (in metres) of wombat sightings/counts according to previously set
134 distance intervals/bands
135
136 * Route_ID: individual ID assigned to each spotlight survey transect
137
138 * Year: year of the spotlight survey (2002–2020)
139
140 * Veg_Group: describes main vegetation group the transect runs through; 11 vegetation groups as
141 defined by Kitchener, A. & Harris, S. (2013). From Forest to Fjaeldmark: Descriptions of
142 Tasmania's Vegetation. Australia: Department of Natural Resources and Environment, Tasmania and
143 used in the TASVEG 4.0 dataset
144
145 * Sightings: number of wombats counted
146
147 * DistBand: For each sighted/counted wombat, perpendicular distance from transect was recorded,
148 using six categories/bands: 0–5, 5–10, 10–20, 20–40, 40–60, and 60–100 m (metres).
149
150 * j,k,i: numerical classifications for 'vegetation group', 'year' and 'distance band/interval',
151 respectively
```

Sometimes description may not be enough

Further research may be required....

■ *What is a 'transect'?*

.
.

Wombat data set
“README” file

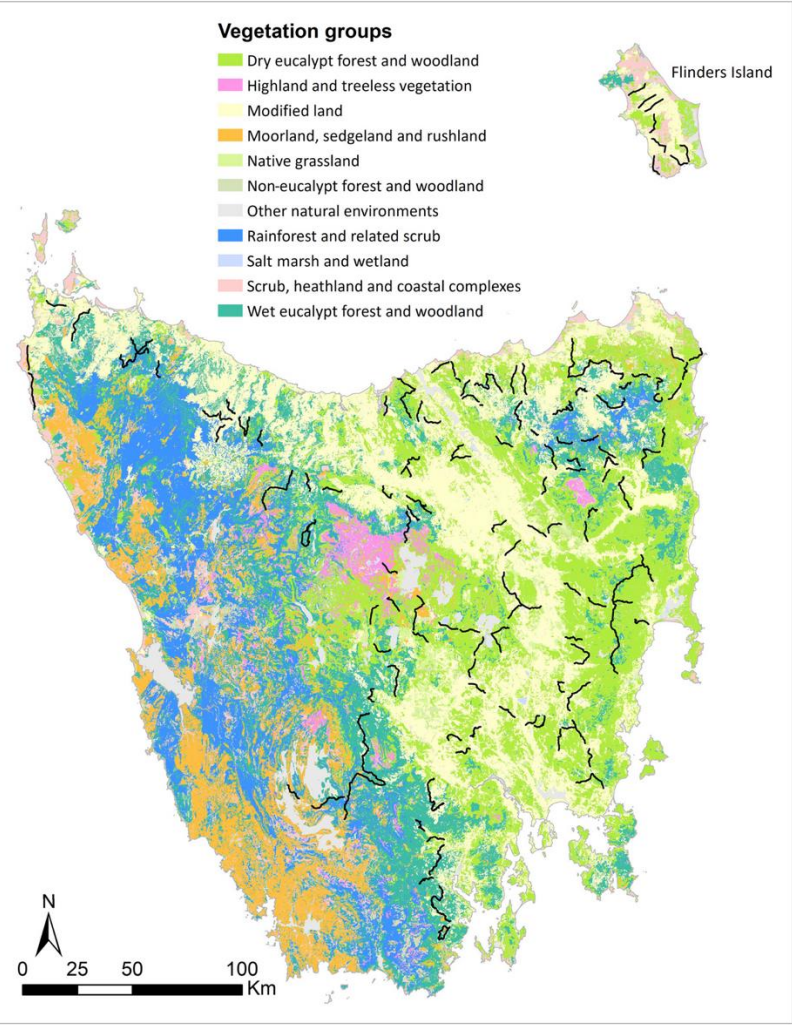


FIGURE 1 [Open in figure viewer](#) [PowerPoint](#)

Tasmania, Australia, with broad groupings of vegetation communities as defined by Kitchener and Harris (2013). Black lines are 180 10-km spotlight transects (172 on the Tasmanian mainland, 8 on Flinders Island), used for annually conducted wildlife surveys by the Department of Natural Resources and Environment (NRE), Tasmania.

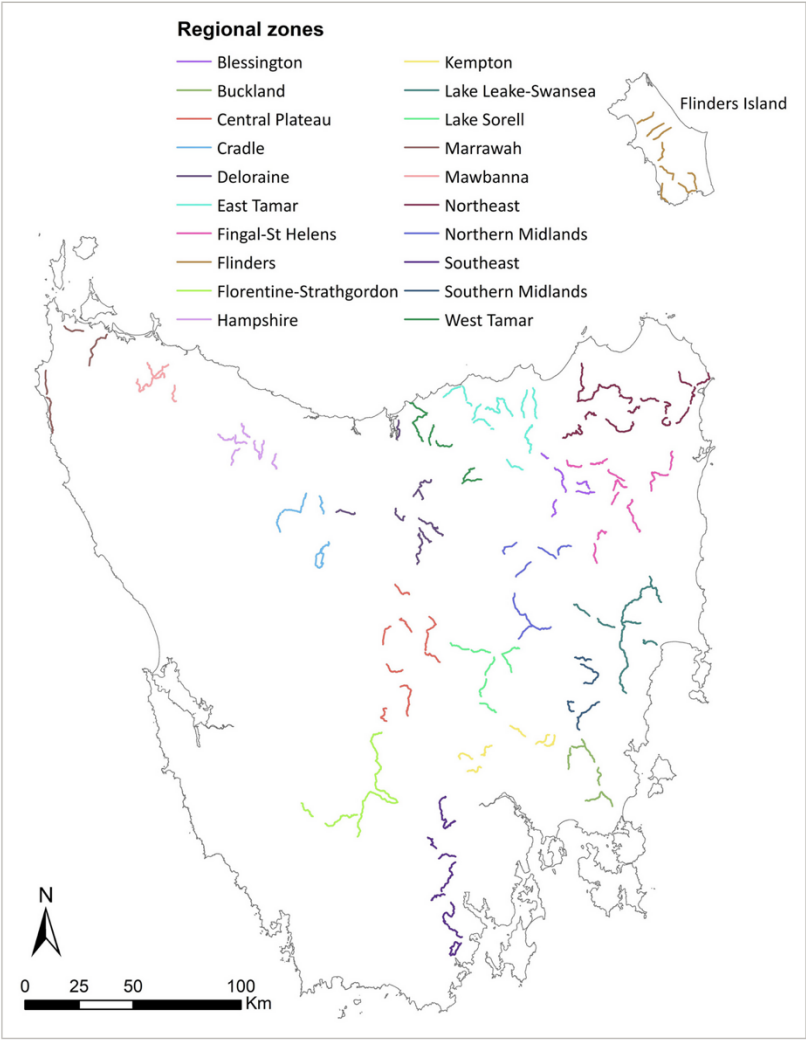


FIGURE 2 [Open in figure viewer](#) [PowerPoint](#)

Tasmania, Australia, with 180 10-km spotlight transects (172 on the Tasmanian mainland, 8 on Flinders Island). On the Tasmanian mainland, bare-nosed wombat (*Vombatus ursinus tasmaniensis*) counts and distance data were recorded from transects spanning 19 regional zones (denoted by transect colour). Flinders Island was treated as single zone comprising the Flinders Island wombat (*Vombatus ursinus ursinus*).

Mainland_habitat_modified.csv*

Missing Data

Rows: 19548 | Columns: 8

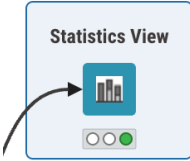
Table Statistics

☐	#	RowID	Column0 <i>Number (integer)</i>	Surveys.Survey_ID <i>Number (integer)</i>	Dist_Min <i>Number (integer)</i>	Dist_Max <i>Number (integer)</i>	Route_ID <i>String</i>	Year <i>Number (integer)</i>	Veg_Group <i>String</i>	Sightings <i>Number (integer)</i>
☐	1	Row0	1	28	0	5	Ansons	2002	Modified land	0
☐	2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
☐	3	Row2	3	28	10	20	?	2002	Modified land	0
☐	4	Row3	4	28	?	40	Ansons-1	2002	Modified land	0
☐	5	Row4	5	28	40	60	Ansons-1	2002	Modified land	?
☐	6	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
☐	7	Row6	7	29	0	5	Ansons-1	2003	Modified land	0
☐	8	Row7	8	29	5	10	Ansons-1	?	Modified land	0
☐	9	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
☐	10	Row9	10	29	20	40	Ansons-1	2003	Modified land	0

• • • • • • • • • •

* Data removed from original file for illustration purposes

Check for missing values



Open in new window

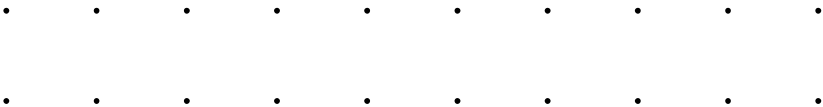
Statistics

Rows: 7 | Columns: 8

Name	# Missing values	Minimum	Maximum	50% Quantile (Median)	Mean	Standard Deviation	Skewness
Surveys.Survey_ID	0	28	7,112	5,317.5	4,936.598	1,716.791	-1.137
Dist_Min	1	0	60	10	22.5	21.165	0.692
Dist_Max	0	5	100	30	39.167	32.968	0.756
Route_ID	1	?	?	?	?	?	?
Year	1	2,002	2,020	2,011	2,010.987	5.483	0.004
Veg_Group	0	?	?	?	?	?	?
Sightings	1	0	15	0	0.239	0.859	5.929

Rows: 19548 | Columns: 8

	#	RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)
☐	1	Row0	1	28	0	5	Ansons-1	2002	Modified land	0
☐	2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
☐	3	Row2	3	28	10	20	?	2002	Modified land	0
☐	4	Row3	4	28	?	40	Ansons-1	2002	Modified land	0
☐	5	Row4	5	28	40	60	Ansons-1	2002	Modified land	?
☐	6	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
☐	7	Row6	7	29	0	5	Ansons-1	2003	Modified land	0
☐	8	Row7	8	29	5	10	Ansons-1	?	Modified land	0
☐	9	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
☐	10	Row9	10	29	20	40	Ansons-1	2003	Modified land	0



Review data dictionary/metadata

Contains detailed information about data

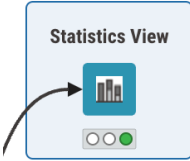
Our missing values are for:

- **Dist_Min:** minimum distance (in metres) of wombat sightings/counts according to previously set distance intervals/bands
- **Route_ID:** individual ID assigned to each spotlight survey transect
- **Year:** year of the spotlight survey (2002-2020)
- **Sightings:** number of wombats counted

```
116 #####
117
118 DATA-SPECIFIC INFORMATION FOR: MAINLAND_HABITAT.csv
119
120 1. Short description: shows various data relating to Tasmanian mainland spotlight surveys
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134 distance intervals/bands
135
136 * Route_ID: individual ID assigned to each spotlight survey transect
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138 * Year: year of the spotlight survey (2002-2020)
139
140 * Veg_Group: describes main vegetation group the transect runs through; 11 vegetation groups as
141 defined by Kitchener, A. & Harris, S. (2013). From Forest to Fjeldmark: Descriptions of
142 Tasmania's Vegetation. Australia: Department of Natural Resources and Environment, Tasmania and
143 used in the TASVEG 4.0 dataset
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145 * Sightings: number of wombats counted
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147 * DistBand: For each sighted/counted wombat, perpendicular distance from transect was recorded,
148 using six categories/bands: 0-5, 5-10, 10-20, 20-40, 40-60, and 60-100 m (metres).
149
150 * j,k,i: numerical classifications for 'vegetation group', 'year' and 'distance band/interval',
151 respectively
```

Wombat data set
"README" file

Check for missing values



Open in new window

Statistics

Rows: 7 | Columns: 8

	Name	# Missing values	Minimum	Maximum	50% Quantile (Median)	Mean	Standard Deviation	Skewness
Nominal	Surveys.Survey_ID	0	28	7,112	5,317.5	4,936.598	1,716.791	-1.137
Ratio	Dist_Min	1	0	60	10	22.5	21.165	0.692
Ratio	Dist_Max	0	5	100	30	39.167	32.968	0.756
Nominal	Route_ID	1	?	?	?	?	?	?
Interval	Year	1	2,002	2,020	2,011	2,010.987	5.483	0.004
Nominal	Veg_Group	0	?	?	?	?	?	?
Ratio	Sightings	1	0	15	0	0.239	0.859	5.929

Rows: 19548 | Columns: 8

	#	RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)
☐	1	Row0	1	28	0	5	Ansons-1	2002	Modified land	0
☐	2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
☐	3	Row2	3	28	10	20	?	2002	Modified land	0
☐	4	Row3	4	28	?	40	Ansons-1	2002	Modified land	0
☐	5	Row4	5	28	40	60	Ansons-1	2002	Modified land	?
☐	6	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
☐	7	Row6	7	29	0	5	Ansons-1	2003	Modified land	0
☐	8	Row7	8	29	5	10	Ansons-1	?	Modified land	0
☐	9	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
☐	10	Row9	10	29	20	40	Ansons-1	2003	Modified land	0

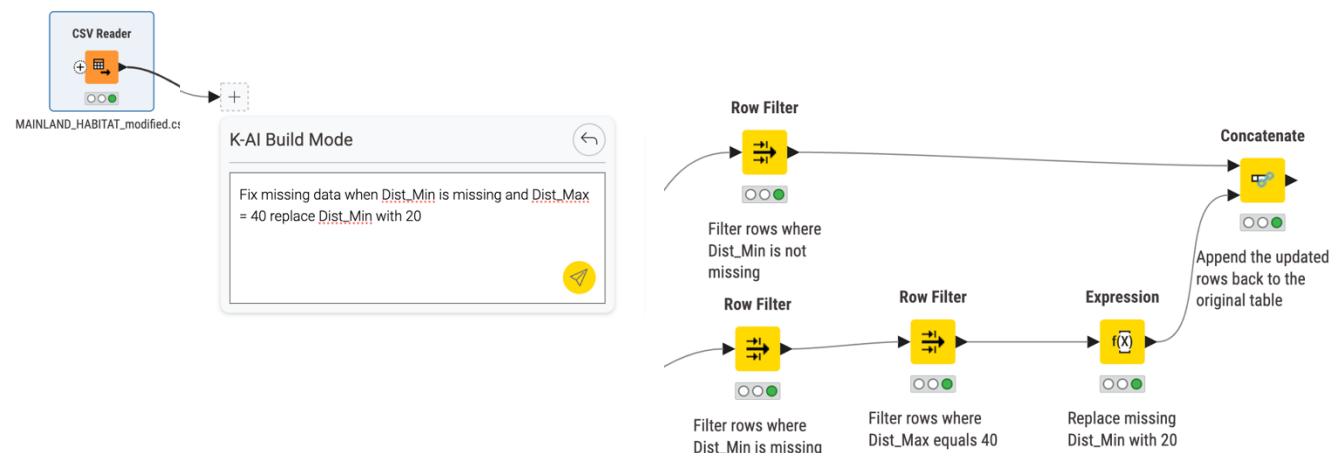
* Dist_Min: minimum distance (in metres) of wombat sightings/counts according to previously set distance intervals/bands

* Dist_Max: maximum distance (in metres) of wombat sightings/counts according to previously set distance intervals/bands

* DistBand: For each sighted/counted wombat, perpendicular distance from transect was recorded, using six categories/bands: 0-5, 5-10, 10-20, 20-40, 40-60, and 60-100 m (metres).

Work out what to do with missing value

- **Dist_Min**: minimum distance (in metres) of wombat sightings/counts according to previously set distance intervals/bands
 - 0-5, 5-10, 10-20, 20-40, 40-60, and 60-100 m
 - Replace value that is missing



RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)
Row0	1	28	0	5
Row1	2	28	5	10
Row2	3	28	10	20
Row3	4	28	20	40
Row4	5	28	40	60
Row5	6	28	60	100
Row6	7	29	0	5
Row7	8	29	5	10
Row8	9	29	10	20
Row9	10	29	20	40

Rows: 19548 | Columns: 8

#	RowID	Column0 Number (integer)	Surveys.Surv... Number (integer)	Dist_Min Number (long)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)
19537	Row19	19538	6890	10	20	Waddamana-7	2019	Dry eucalypt forest	0
19538	Row19	19539	6890	20	40	Waddamana-7	2019	Dry eucalypt forest	0
19539	Row19	19540	6890	40	60	Waddamana-7	2019	Dry eucalypt forest	0
19540	Row19	19541	6890	60	100	Waddamana-7	2019	Dry eucalypt forest	0
19541	Row19	19542	7076	0	5	Waddamana-7	2020	Dry eucalypt forest	1
19542	Row19	19543	7076	5	10	Waddamana-7	2020	Dry eucalypt forest	2
19543	Row19	19544	7076	10	20	Waddamana-7	2020	Dry eucalypt forest	0
19544	Row19	19545	7076	20	40	Waddamana-7	2020	Dry eucalypt forest	0
19545	Row19	19546	7076	40	60	Waddamana-7	2020	Dry eucalypt forest	0
19546	Row19	19547	7076	60	100	Waddamana-7	2020	Dry eucalypt forest	0
19547	Row19	19548	28	20	40	Ansons-1	2002	Modified land	0
19548	Row3	4	28	20	40	Ansons-1	2002	Modified land	0

Concatenate: Appends to bottom of data set

Work out what to do with missing values: Interpolation

- **Year:** year of the spotlight survey (2002-2020)
 - Missing -> delete row
 - Missing -> take value in front/behind (check not on year boundary)
- **Route_ID:** individual ID assigned to each spotlight survey transect
 - Missing -> delete row
 - Missing -> take value in front/behind (check not on route boundary)

1: Concatenated table

Rows: 19548 | Columns: 8

#	RowID	Column0	Surveys.Surv...	Dist_Min	Dist_Max	Route_ID	Year	Veg_Group	Sightings
		Number (integer)	Number (integer)	Number (long)	Number (integer)	String	Number (integer)	String	Number (integer)
1	Row0	1	28	0	5	Ansons-1	2002	Modified land	0
2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
3	Row2	3	28	10	20	Ansons-1	2002	Modified land	0
4	Row4	5	28	40	60	Ansons-1	2002	Modified land	0
5	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
6	Row6	7	29	0	5	Ansons-1	2003	Modified land	0
7	Row7	8	29	5	10	Ansons-1	2003	Modified land	0
8	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
9	Row9	10	29	20	40	Ansons-1	2003	Modified land	0
10	Row10	11	29	40	60	Ansons-1	2003	Modified land	0

Rows: 19546 | Columns: 8

#	RowID	Column0	Surveys.Surv...	Dist_Min	Dist_Max	Route_ID	Year	Veg_Group	Sightings
		Number (integer)	Number (integer)	Number (long)	Number (integer)	String	Number (integer)	String	Number (integer)
1	Row0	1	28	0	5	Ansons-1	2002	Modified land	0
2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
3	Row4	5	28	40	60	Ansons-1	2002	Modified land	0
4	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
5	Row6	7	29	0	5	Ansons-1	2003	Modified land	0
6	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
7	Row9	10	29	20	40	Ansons-1	2003	Modified land	0
8	Row10	11	29	40	60	Ansons-1	2003	Modified land	0
9	Row11	12	29	60	100	Ansons-1	2003	Modified land	0
10	Row12	13	30	0	5	Ansons-1	2004	Modified land	1

Delete row

Work out what to do with missing values: Interpolation

- **Year:** year of the spotlight survey (2002-2020)
 - Missing -> delete row
 - Missing -> **nearest neighbour** (take value in front/behind (check not on year boundary))
- **Route_ID:** individual ID assigned to each spotlight survey transect
 - Missing -> delete row
 - Missing -> **nearest neighbour** (take value in front/behind (check not on route boundary))

1: Concatenated table

Rows: 19548 | Columns: 8

#	RowID	Column0	Surveys.Surv...	Dist_Min	Dist_Max	Route_ID	Year	Veg_Group	Sightings
		Number (integer)	Number (integer)	Number (long)	Number (integer)	String	Number (integer)	String	Number (integer)
1	Row0	1	28	0	5	Ansons-1	2002	Modified land	0
2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
3	Row2	3	28	10	20	Ansons-1	2002	Modified land	0
4	Row4	5	28	40	60	Ansons-1	2002	Modified land	0
5	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
6	Row6	7	29	0	5	Ansons-1	2002	Modified land	0
7	Row7	8	29	5	10	Ansons-1	2002	Modified land	0
8	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
9	Row9	10	29	20	40	Ansons-1	2003	Modified land	0
10	Row10	11	29	40	60	Ansons-1	2003	Modified land	0

Rows: 19548 | Columns: 8

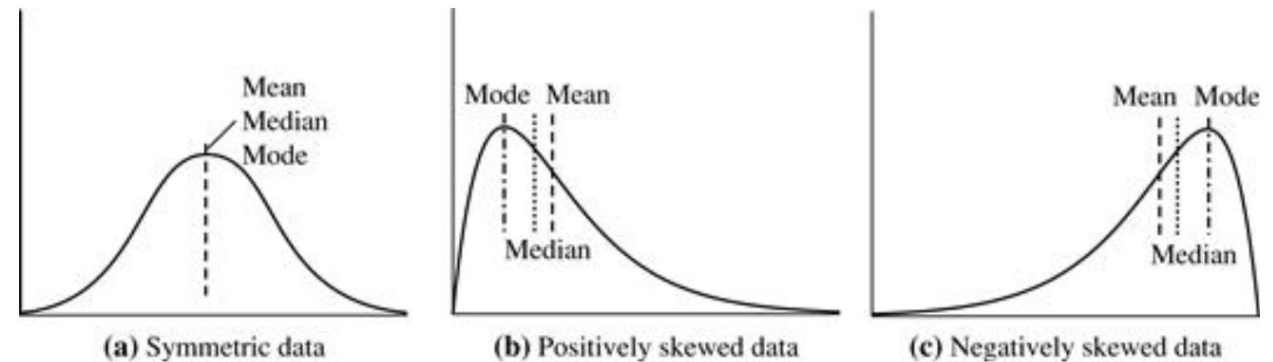
#	RowID	Column0	Surveys.Surv...	Dist_Min	Dist_Max	Route_ID	Year	Veg_Group	Sightings
		Number (integer)	Number (integer)	Number (long)	Number (integer)	String	Number (integer)	String	Number (integer)
1	Row0	1	28	0	5	Ansons-1	2002	Modified land	0
2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
3	Row2	3	28	10	20	Ansons-1	2002	Modified land	0
4	Row4	5	28	40	60	Ansons-1	2002	Modified land	0
5	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
6	Row6	7	29	0	5	Ansons-1	2003	Modified land	0
7	Row7	8	29	5	10	Ansons-1	2003	Modified land	0
8	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
9	Row9	10	29	20	40	Ansons-1	2003	Modified land	0
10	Row10	11	29	40	60	Ansons-1	2003	Modified land	0

Interpolate row

Work out what to do with missing values: Interpolation

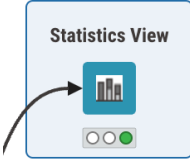
Sightings: number of wombats counted (ratio)

- Replace with measure of central tendency (quantitative data)
 - Mean
 - Median (when skewedness is
 - greater than 1 or
 - less than -1)
- Replace with global constant (e.g., 'unknown', 'NA' etc)
- Remove tuple (all data in row)



• • • • • • • • • •
• • • • • • • • • •

Check for missing values



Open in new window

Statistics

Rows: 7 | Columns: 8

	Name	# Missing values	Minimum	Maximum	50% Quantile (Median)	Mean	Standard Deviation	Skewness
Nominal	Surveys.Survey_ID	0	28	7,112	5,317.5	4,936.598	1,716.791	-1.137
Ratio	Dist_Min	1	0	60	10	22.5	21.165	0.692
Ratio	Dist_Max	0	5	100	30	39.167	32.968	0.756
Nominal	Route_ID	1	?	?	?	?	?	?
Interval	Year	1	2,002	2,020	2,011	2,010.987	5.483	0.004
Nominal	Veg_Group	0	?	?	?	?	?	?
Ratio	Sightings	1	0	15	0	0.239	0.859	5.929

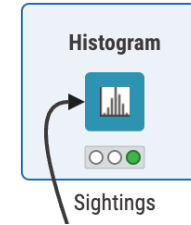
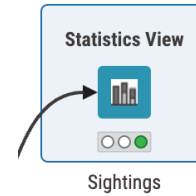
Rows: 19548 | Columns: 8

	#	RowID	Column0 Number (integer)	Surveys.Survey_ID Number (integer)	Dist_Min Number (integer)	Dist_Max Number (integer)	Route_ID String	Year Number (integer)	Veg_Group String	Sightings Number (integer)
☐	1	Row0	1	28	0	5	Ansons-1	2002	Modified land	0
☐	2	Row1	2	28	5	10	Ansons-1	2002	Modified land	0
☐	3	Row2	3	28	10	20	?	2002	Modified land	0
☐	4	Row3	4	28	?	40	Ansons-1	2002	Modified land	0
☐	5	Row4	5	28	40	60	Ansons-1	2002	Modified land	?
☐	6	Row5	6	28	60	100	Ansons-1	2002	Modified land	0
☐	7	Row6	7	29	0	5	Ansons-1	2003	Modified land	0
☐	8	Row7	8	29	5	10	Ansons-1	?	Modified land	0
☐	9	Row8	9	29	10	20	Ansons-1	2003	Modified land	0
☐	10	Row9	10	29	20	40	Ansons-1	2003	Modified land	0



Work out what to do with missing values: Interpolation

- Sightings
 - Check distribution (highly skewed)



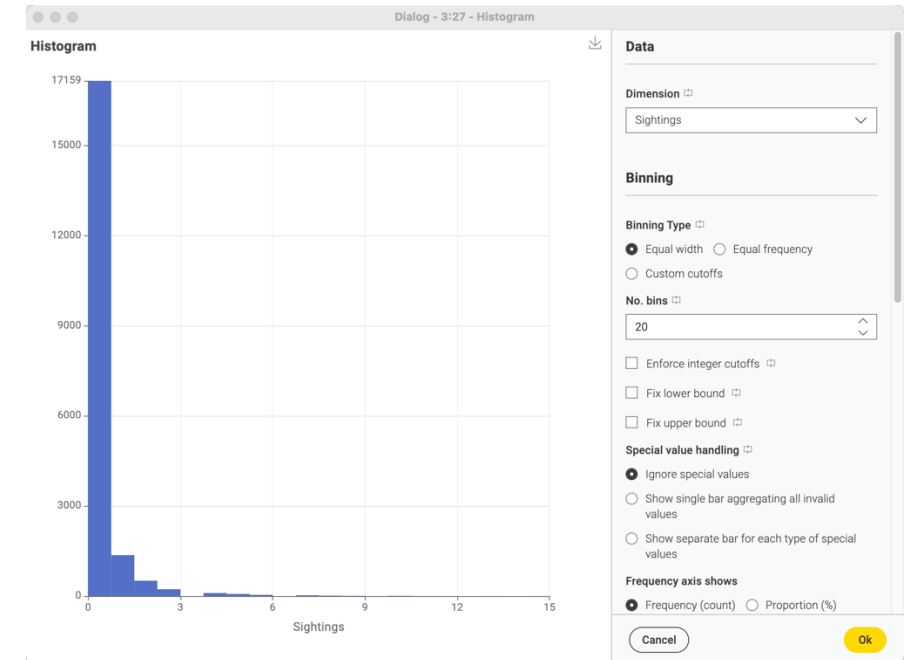
Statistics

[Open in new window](#)

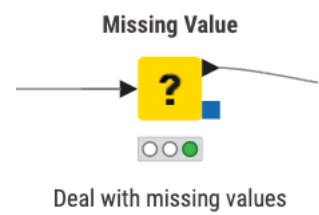
Rows: 1 | Columns: 7

Name	Minimum	Maximum	50% Quantile (Median)	Mean	Standard Deviation	Skew
Sightings	0	15	0	0.239	0.859	5.929

- Replace with Median (not mean)



KNIME Missing Values



Dialog - 3:6 - Missing Value (Deal with missing values)

Default Column Settings Flow Variables Job Manager Selection Memory Policy

Column Search

Filter Options

None

Column0
Surveys.Survey_ID
Dist_Min
Dist_Max
Route_ID
Year
Veg_Group
Sightings

Year

Remove

Previous Value*

Route_ID

Remove

Previous Value*

Sightings

Remove

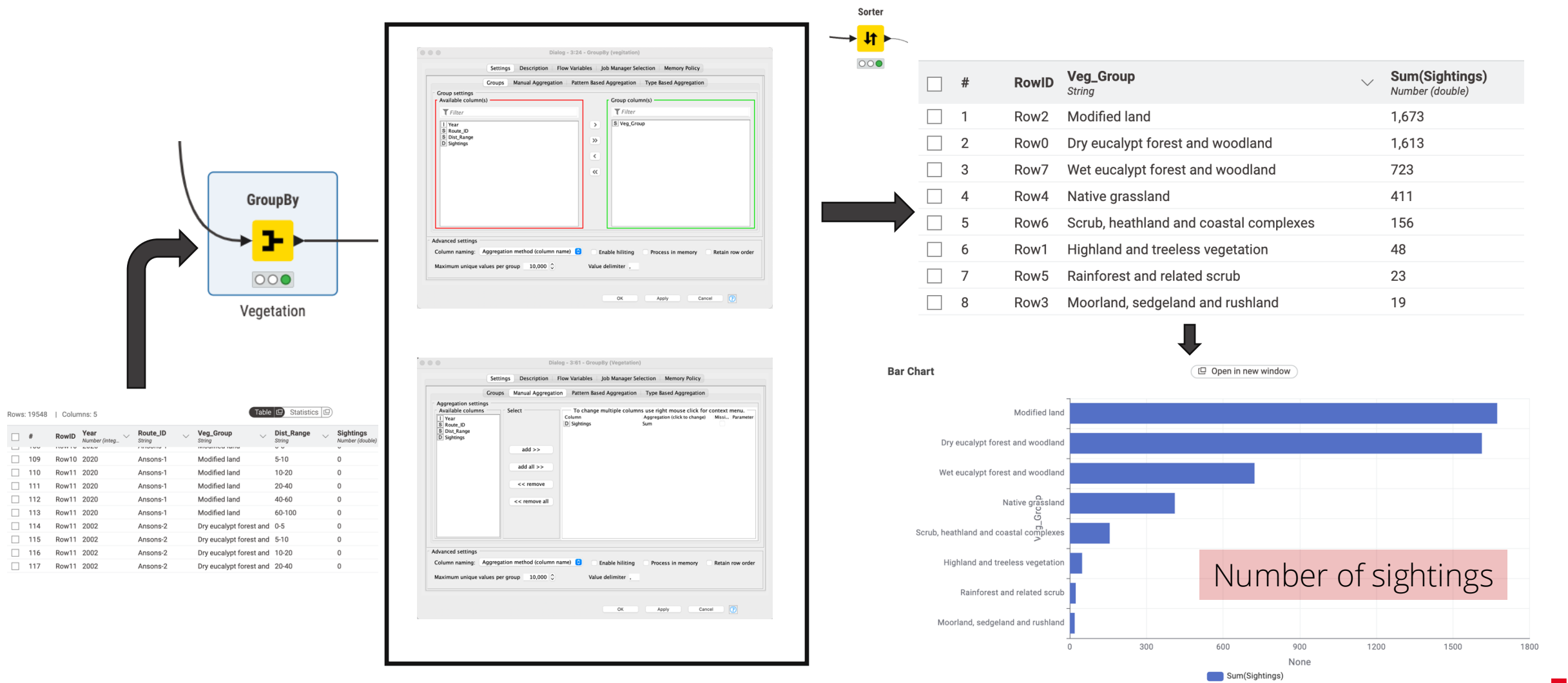
Median

Add

OK Apply Cancel ?

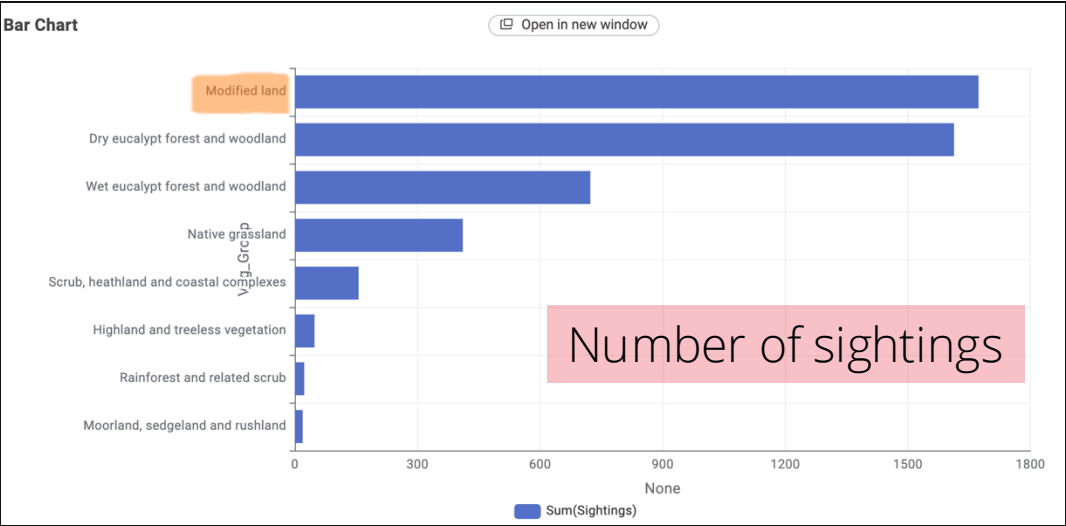
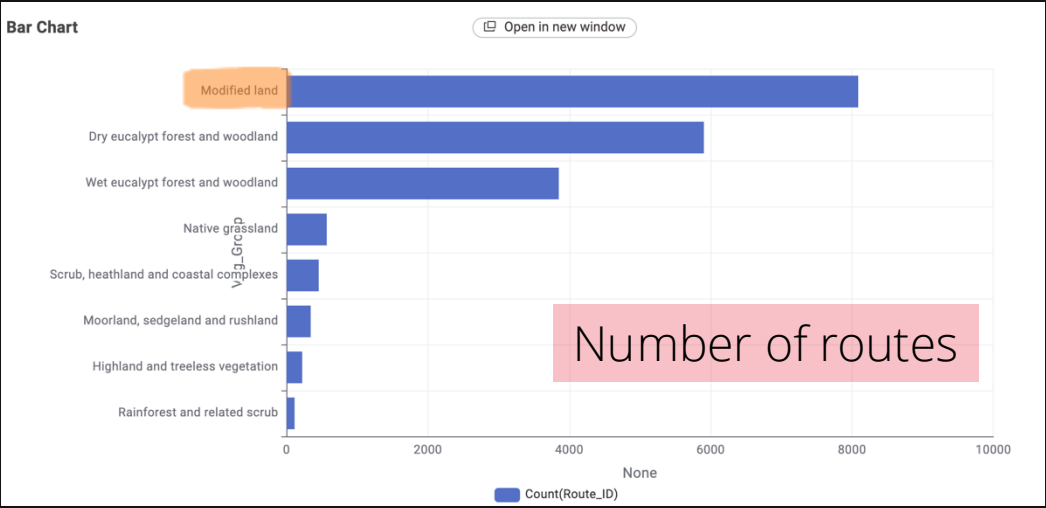
The dialog box is titled 'Dialog - 3:6 - Missing Value (Deal with missing values)'. It has tabs for 'Default', 'Column Settings', 'Flow Variables', 'Job Manager Selection', and 'Memory Policy'. The 'Column Settings' tab is active. On the left, there is a 'Column Search' section with a 'Filter Options' dropdown set to 'None'. Below this is a list of columns: 'Column0', 'Surveys.Survey_ID', 'Dist_Min', 'Dist_Max', 'Route_ID' (highlighted in blue), 'Year', 'Veg_Group', and 'Sightings'. On the right, there are three rows corresponding to the selected columns. Each row has a 'Remove' button and a dropdown menu for the replacement value. For 'Year', the dropdown is 'Previous Value*'. For 'Route_ID', the dropdown is 'Previous Value*'. For 'Sightings', the dropdown is 'Median'. At the bottom left is an 'Add' button. At the bottom right are 'OK', 'Apply', and 'Cancel' buttons, along with a help icon.

What type of vegetation do wombats like?



How many surveys taken at per vegetation site?

☐	#	RowID	Veg_Group String	Sum(Sightings) Number (double)	Count(Route_ID) Number (integer)
☐	1	Row2	Modified land	1,673	8088
☐	2	Row0	Dry eucalypt forest and woodland	1,613	5904
☐	3	Row7	Wet eucalypt forest and woodland	723	3852
☐	4	Row4	Native grassland	411	570
☐	5	Row6	Scrub, heathland and coastal complexes	156	456
☐	6	Row3	Moorland, sedgeland and rushland	19	342
☐	7	Row1	Highland and treeless vegetation	48	222
☐	8	Row5	Rainforest and related scrub	23	114



How many wombat sightings per vegetation type?

Transformation: Number of wombat sightings per vegetation type

- wombat sightings/number of vegetation surveys

☐	#	RowID	Veg_Group String	Sum(Sightings) ↓ Number (double)	Count(Route_ID) ↓ Number (integer)	Sum_by_Count Number (double)
☐	1	Row2	Modified land	1,673	8088	0.207
☐	2	Row0	Dry eucalypt forest and woodland	1,613	5904	0.273
☐	3	Row7	Wet eucalypt forest and woodland	723	3852	0.188
☐	4	Row4	Native grassland	411	570	0.721
☐	5	Row6	Scrub, heathland and coastal complex	156	456	0.342
☐	7	Row1	Highland and treeless vegetation	48	222	0.216
☐	8	Row5	Rainforest and related scrub	23	114	0.202
☐	6	Row3	Moorland, sedgeland and rushland	19	342	0.056

Expression



Perform the division of Sum(Sightings) by Count(Route_ID) and create a new column

Input table

ROW_NUMBER	INTEGER
ROW_INDEX	INTEGER
ROW_ID	STRING
Veg_Group	STRING
Sum(Sightings)	FLDAT
Count(Route_ID)	INTEGER

Flow variables

Home workspace	STRING
----------------	--------

Expression 1

```
1 | Sum(Sightings) / Count(Route_ID)
```

Output creates "Sum_by_Count"

Append Replace

Sum_by_Count

Add expression

Ask KAI

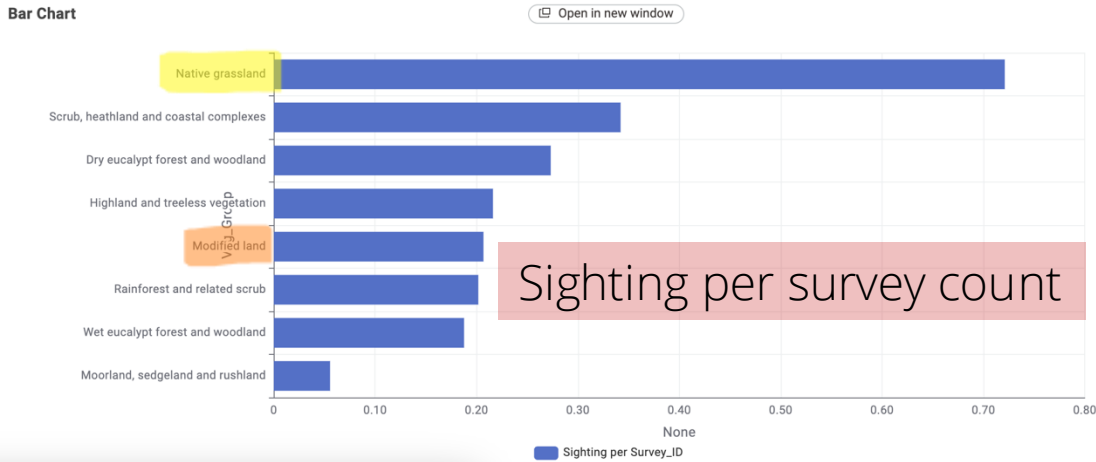
Evaluate first 10 rows

Output preview

To see the preview, evaluate the expression using the button above.

Cancel

OK



Lot more 'rainforest and related scrub' than in survey

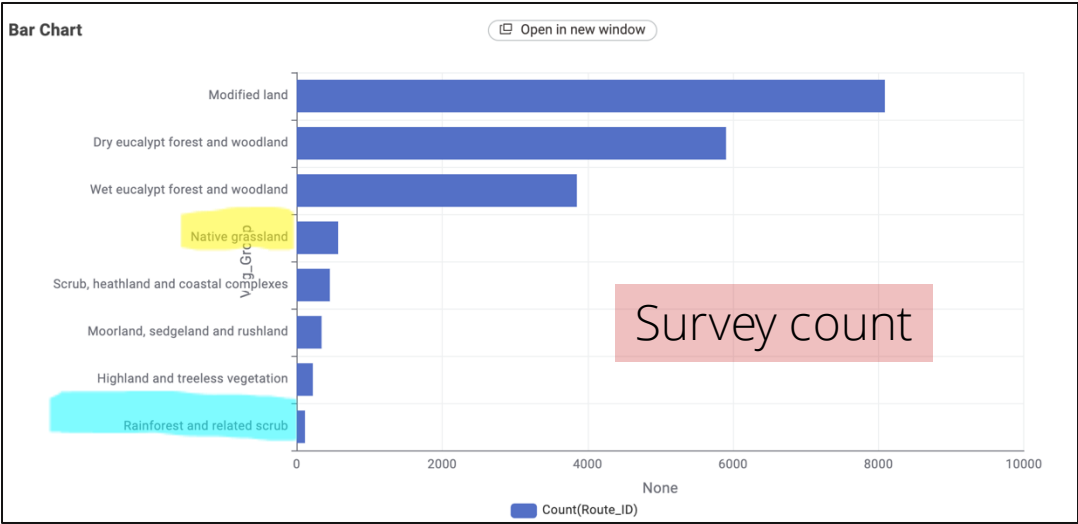
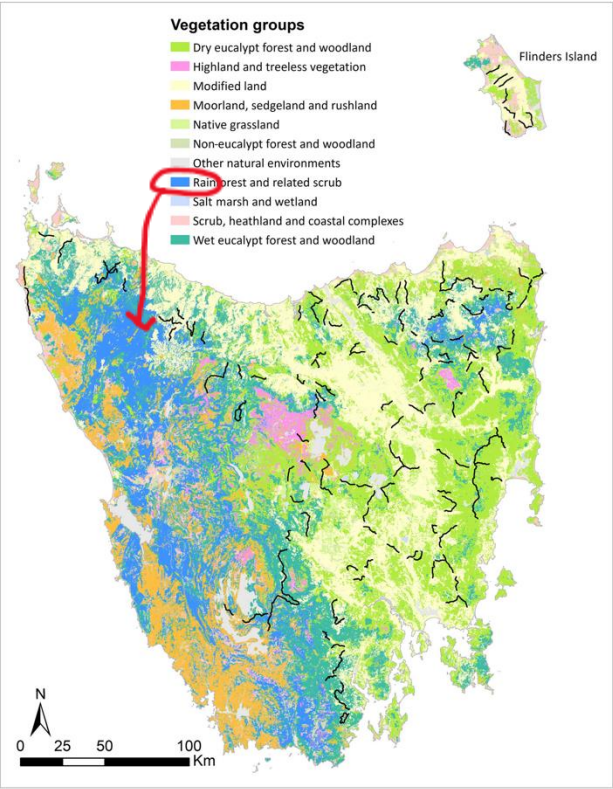


FIGURE 1 [Open in figure viewer](#) [PowerPoint](#)

Tasmania, Australia, with broad groupings of vegetation communities as defined by Kitchener and Harris (2013). Black lines are 180 10-km spotlight transects (172 on the Tasmanian mainland, 8 on Flinders Island), used for annually conducted wildlife surveys by the Department of Natural Resources and Environment (NRE), Tasmania.

Tasmania percent vegetation coverage

Rows: 11 | Columns: 4

Table Statistics

☐	#	RowID	Vegetation Group <i>String</i>	Area (sqm) <i>Number (double)</i>	Area (ha) <i>Number (double)</i>	Area (%) <i>Number (double)</i>
☐	1	Row0	Dry eucalypt forest and woodland	15,150,630,522.64	1,515,063.052	23.09
☐	2	Row2	Modified land	15,104,899,663.79	1,510,489.966	23.02
☐	3	Row10	Wet eucalypt forest and woodland	11,147,169,437.13	1,114,716.944	16.989
☐	4	Row7	Rainforest and related scrub	7,570,276,980.26	757,027.698	11.537
☐	5	Row3	Moorland, sedgeland and rushland	5,966,284,708.69	596,628.471	9.093
☐	6	Row9	Scrub, heathland and coastal complexes	3,631,370,090.48	363,137.009	5.534
☐	7	Row6	Other natural environments	2,399,509,441.63	239,950.944	3.657
☐	8	Row5	Non eucalypt forest and woodland	1,792,588,725.23	179,258.873	2.732
☐	9	Row1	Highland and treeless vegetation	1,355,493,858.57	135,549.386	2.066
☐	10	Row4	Native grassland	1,211,297,634.44	121,129.763	1.846
☐	11	Row8	Saltmarsh and wetland	286,001,004.31	28,600.1	0.436

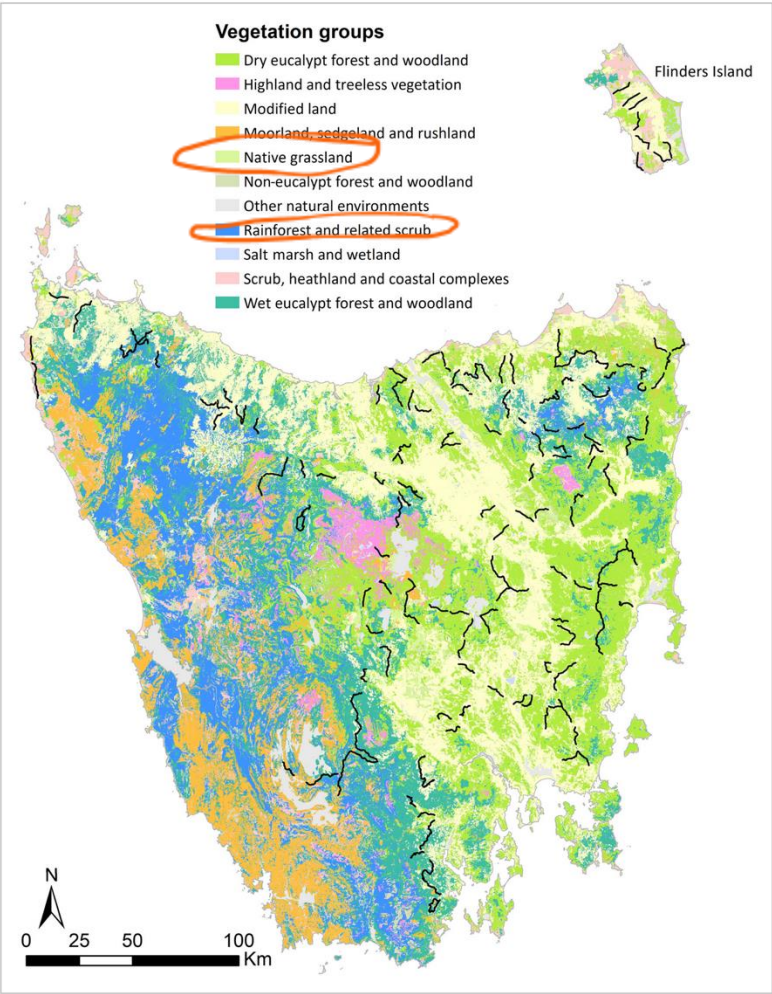
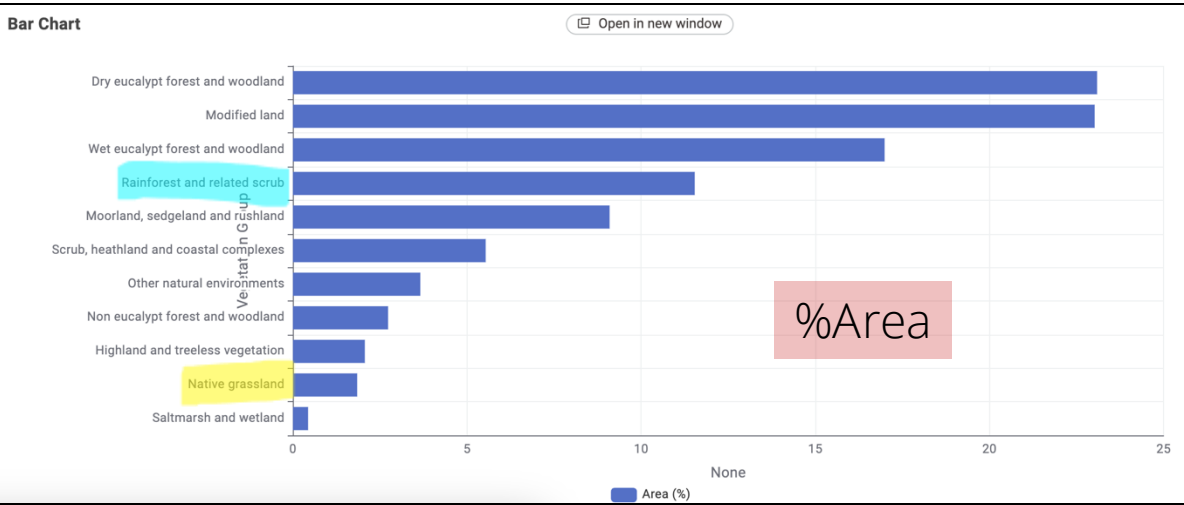


FIGURE 1

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Tasmania, Australia, with broad groupings of vegetation communities as defined by Kitchener and Harris (2013). Black lines are 180 10-km spotlight transects (172 on the Tasmanian mainland, 8 on Flinders Island), used for annually conducted wildlife surveys by the Department of Natural Resources and Environment (NRE), Tasmania.

Lot more 'rainforest and related scrub' than in survey

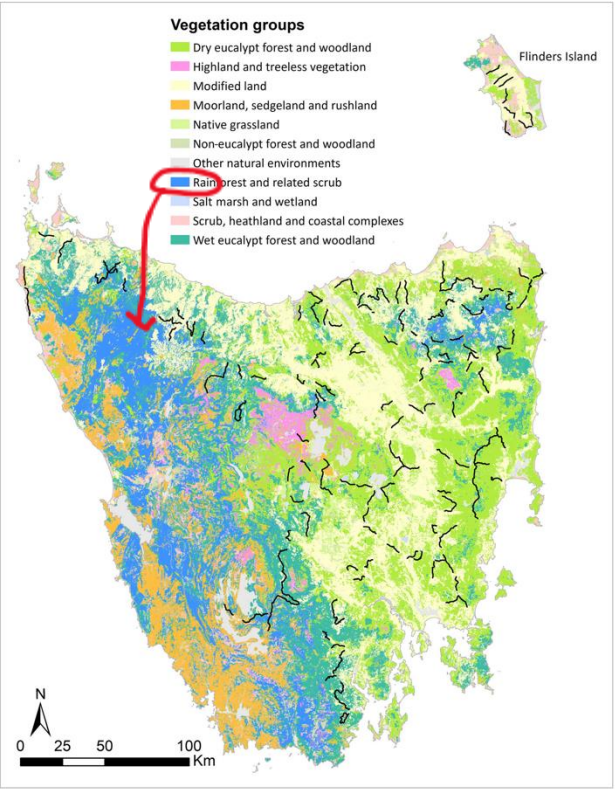
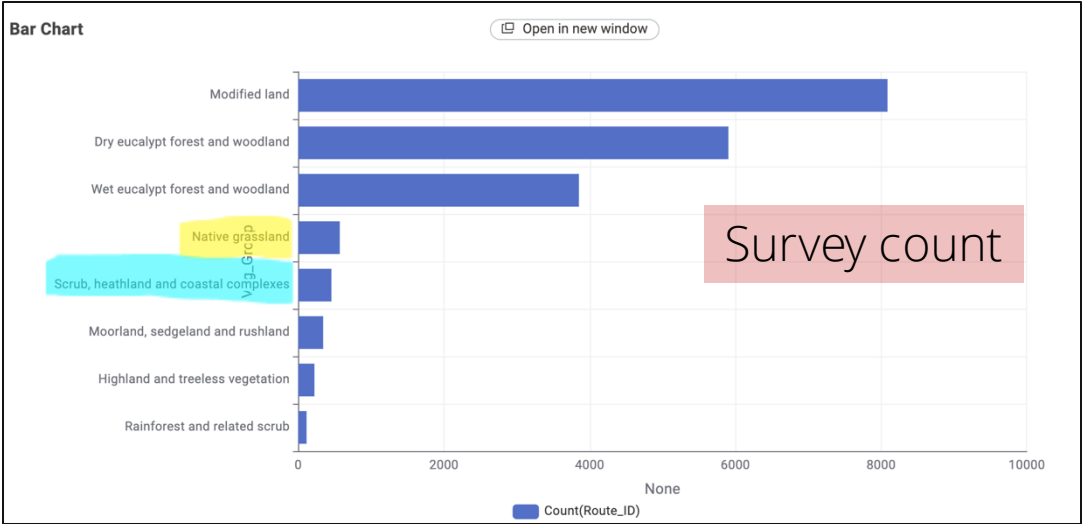
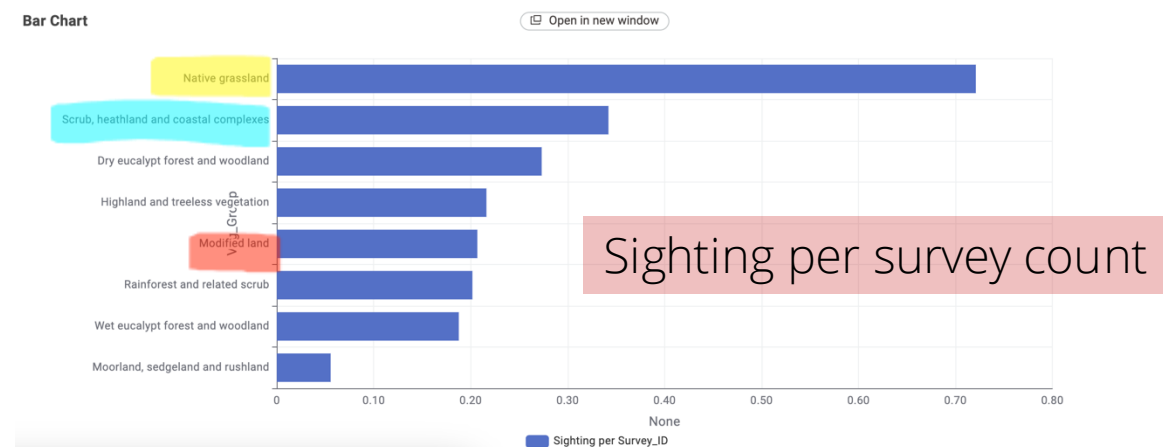
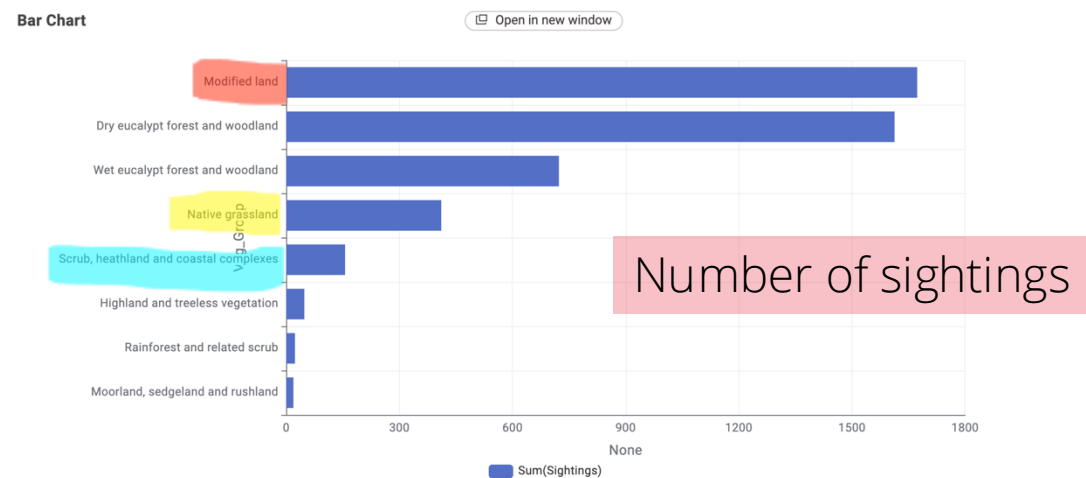


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Tasmania, Australia, with broad groupings of vegetation communities as defined by Kitchener and Harris (2013). Black lines are 180 10-km spotlight transects (172 on the Tasmanian mainland, 8 on Flinders Island), used for annually conducted wildlife surveys by the Department of Natural Resources and Environment (NRE), Tasmania.



What habitat should be protected?



<https://nre.tas.gov.au/wildlife-management/fauna-of-tasmania/mammals/possums-kangaroos-and-wombats/wombat>

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For much more complicated story see:

Knoblauch W, Carver S, Driessen MM, Gales R, & Richards SA (2023). Abundance and population growth estimates for bare-nosed wombats. Ecology and Evolution 13.

<https://onlinelibrary.wiley.com/doi/full/10.1002/ece3.10465>

Exercise1.0 Data Processing with KNIME (follow Canvas guide)

Scenario

Imagine you have been asked to create a set of visualisations that allows users to explore the energy consumption of televisions currently available on the Australian market.

Overview

1. Find the data set
2. Open data set in KNIME
3. Become familiar with data set
4. Clean the data set
5. Transform the data set

